

Great work.

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Section 7.2

9a. $P(|\bar{Y} - \mu| \leq .3) = P(-.3 \leq \bar{Y} - \mu \leq .3)$ ✓
 $= P\left(\frac{-.3}{2/\sqrt{16}} \leq \frac{\bar{Y} - \mu}{2/\sqrt{16}} \leq \frac{.3}{2/\sqrt{16}}\right)$ ✓
 $= P\left(\frac{-.6}{5} \leq Z \leq \frac{.6}{5}\right)$ ✓
 $= \text{normalcdf}(-.6/5, .6/5, 0, 1)$ ✓
 ≈ 0.7698 ✓

- b For $n=25$, lower Bound = -1.5
For $n=36$, lower Bound = -1.8
For $n=49$, lower Bound = -2.1
For $n=64$, lower Bound = -2.4

Ticahilities:

$\Rightarrow \text{normalcdf}(-1.5, 1.5, 0, 1) \approx .8664$ ✓
 $\Rightarrow \text{normalcdf}(-1.8, 1.8, 0, 1) \approx .9282$ ✓
 $\Rightarrow \text{normalcdf}(-2.1, 2.1, 0, 1) \approx .9642$ ✓
 $\Rightarrow \text{normalcdf}(-2.4, 2.4, 0, 1) \approx .9836$ ✓

- c. It increases as n increases ✓

11. Let $\bar{Y}$ = avg. basal area of pine trees. $Y(\mu, 4/\sqrt{9})$
 $P(|\bar{Y} - \mu| \leq 2) = P(-2 \leq \bar{Y} - \mu \leq 2)$ ✓
 $= P\left(\frac{-2 \cdot 3}{4} \leq \frac{\bar{Y} - \mu}{4/\sqrt{9}} \leq \frac{2 \cdot 3}{4}\right)$ ✓
 $= P(-1.5 \leq Z \leq 1.5)$ ✓
 $\approx .8664$ (from b above) ✓

12 Find n such that $P(|\bar{Y} - \mu| \leq 1) = .9$
 $\Rightarrow P(-1 \leq \bar{Y} - \mu \leq 1) = .9$ ✓
 $P\left(\frac{-1 \cdot \sqrt{n}}{4} \leq \frac{\bar{Y} - \mu}{4/\sqrt{n}} \leq \frac{1 \cdot \sqrt{n}}{4}\right) = .9$

$$P(-2 \leq Z \leq 2) = .9$$

$$z = \text{invNorm}(.95, 0, 1) \approx 1.645$$

$$\frac{\sqrt{n}}{4} = 1.645$$

$$n \approx 43.2964$$

$$\text{use } n = 44$$

13. Let $\bar{Y}$ = avg. amount of LCB0. $\bar{Y} \sim N(\mu, \sigma/\sqrt{10})$

$$P(|\bar{Y} - \mu| \leq .5) = P(-.5 \leq \bar{Y} - \mu \leq .5)$$

$$= P\left(\frac{-.5 \cdot 5}{1} \leq \frac{\bar{Y} - \mu}{1/5} \leq \frac{.5 \cdot 5}{1}\right)$$

$$= P(-2.5 \leq Z \leq 2.5)$$

$$= \text{normalcdf}(-2.5, 2.5, 0, 1)$$

$$\approx .9876$$

15 $E(\bar{X} - \bar{Y}) = E(\bar{X}) - E(\bar{Y}) = \mu_1 - \mu_2$

b. $V(\bar{X} - \bar{Y}) = V(\bar{X}) + V(\bar{Y}) = \frac{\sigma_1^2}{n} + \frac{\sigma_2^2}{n} = \frac{1 + 4}{nn}$
independent samples

c. $P(|(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)| \leq 1) = .95$

$$= P(-1 \leq \bar{X} - \bar{Y} - (\mu_1 - \mu_2) \leq 1) = .95$$

$$= P\left(\frac{-1}{\sqrt{\frac{1 + 4}{nn}}} \leq \frac{\bar{X} - \bar{Y} - (\mu_1 - \mu_2)}{\sqrt{\frac{1 + 4}{nn}}} \leq \frac{1}{\sqrt{\frac{1 + 4}{nn}}}\right) = .95$$

$$= P\left(\frac{-1}{\sqrt{4 \cdot 4.5}} \leq Z \leq \frac{1}{\sqrt{4 \cdot 4.5}}\right) = .95$$

$$= P\left(-\sqrt{\frac{n}{4.5}} \leq Z \leq \sqrt{\frac{n}{4.5}}\right) = .95$$

$$\sqrt{\frac{n}{4.5}} = \text{invNorm}(.975, 0, 1) \approx 1.96$$

$$n \approx [\sqrt{4.5} \cdot 1.96]^2 = 27.2872$$

$$\text{use } n = 28$$

Nice.

16. Let $\bar{A}, \bar{B}$ denote avg. amount of LC50 for species A, B

$$\begin{aligned}
 P(\bar{A} - \bar{B} \geq 1) &= P(\bar{A} - \bar{B} - (\mu_A - \mu_B) \geq 1 - (\mu_A - \mu_B)) \\
 &= P\left(\frac{\bar{A} - \bar{B} - (\mu_A - \mu_B)}{\sqrt{10 \cdot 1.2}} \geq \frac{1 - (\mu_A - \mu_B)}{\sqrt{10 \cdot 1.2}}\right) \\
 &= P(Z \geq 2.88675) \checkmark \\
 &= \text{normalcdf}(2.88675, 1E99, 0, 1) \checkmark \\
 &\approx .0019 \checkmark
 \end{aligned}$$

20. Suppose $U \sim \text{chisqvar}(\nu)$

a. $E(U) = \nu$

ii. $V(U) = 2\nu$ $\checkmark$

b. $E\left(\frac{(n-1)S^2}{\sigma^2}\right) = n-1$

$$\frac{n-1}{\sigma^2} E(S^2) = n-1$$

$$E(S^2) = \sigma^2 \checkmark$$

ii. $V\left(\frac{(n-1)S^2}{\sigma^2}\right) = 2(n-1)$

$$\frac{(n-1)^2}{\sigma^4} V(S^2) = 2(n-1)$$

$$V(S^2) = \frac{2\sigma^4}{n-1} \checkmark$$